MACHINE LEARNING ANALYSIS

Campaign Finance Donor Value Prediction

How much do we ask for?

Ridge Regression | 423,576 Donors | $R^2 = 0.974$ | Median Error = $62

NYC Campaign Finance Board | 2013–2021 Elections | 760,955 Transactions

EXECUTIVE SUMMARY

Key findings from the NYC Campaign Finance donor value prediction model

760,955

Total Transactions

Across 2013, 2017 & 2021 NYC elections

150K+

Unique Donors

Identified via name + ZIP fingerprint

00

Median Donor Total

Heavy-tailed: mean is 99

72.6%

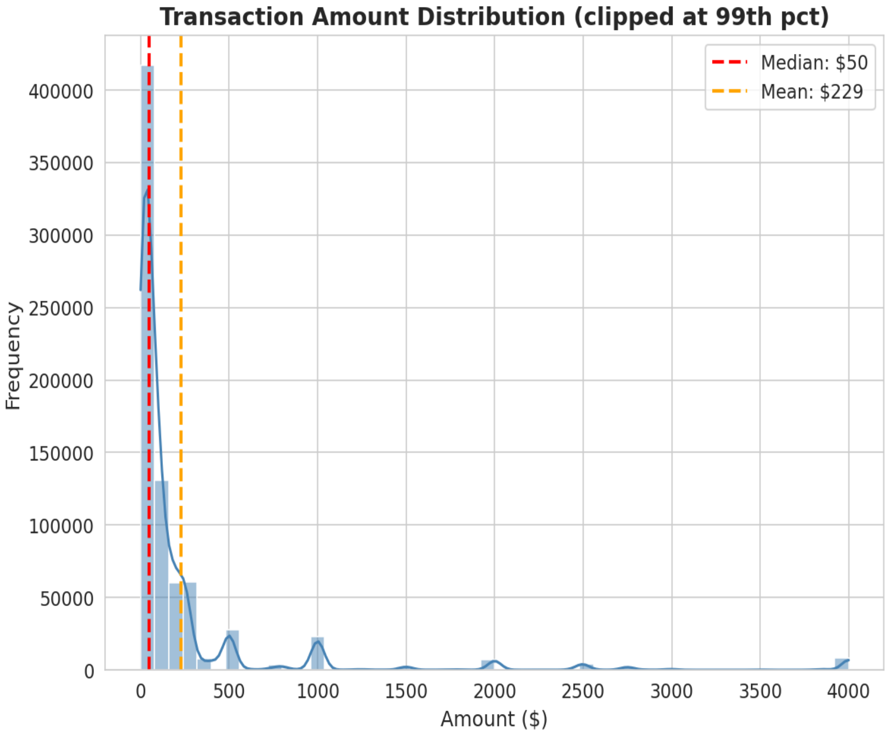
Single-Gift Donors

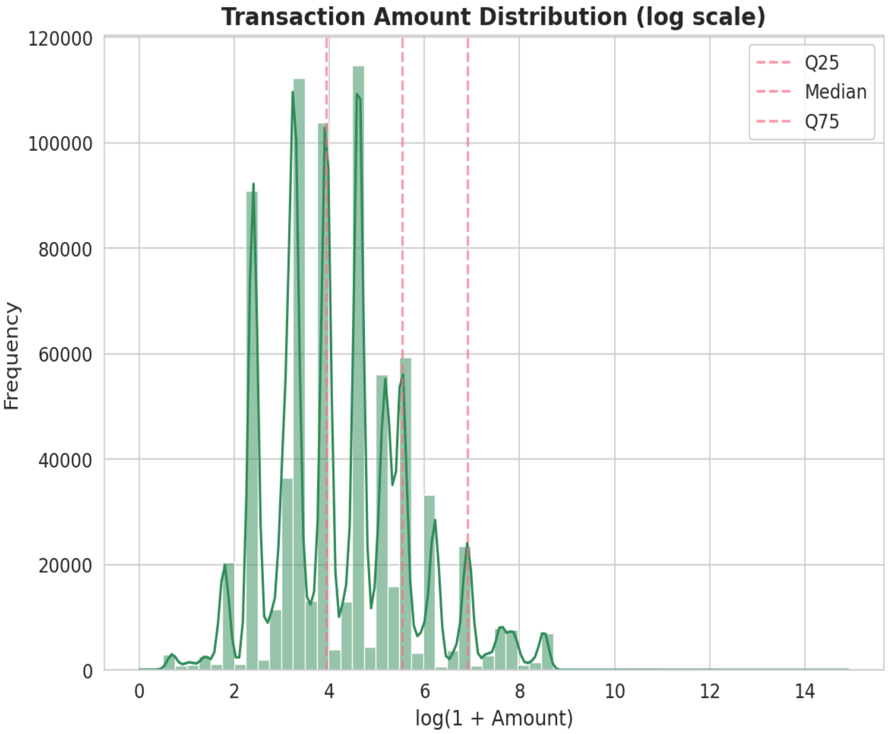
Most donors contribute only once

Model $R^2 = 0.221$ | Median Dollar Error = 2 | Ridge Regression on log-transformed target | Permutation importance used for explainability

TRANSACTION AMOUNT DISTRIBUTION

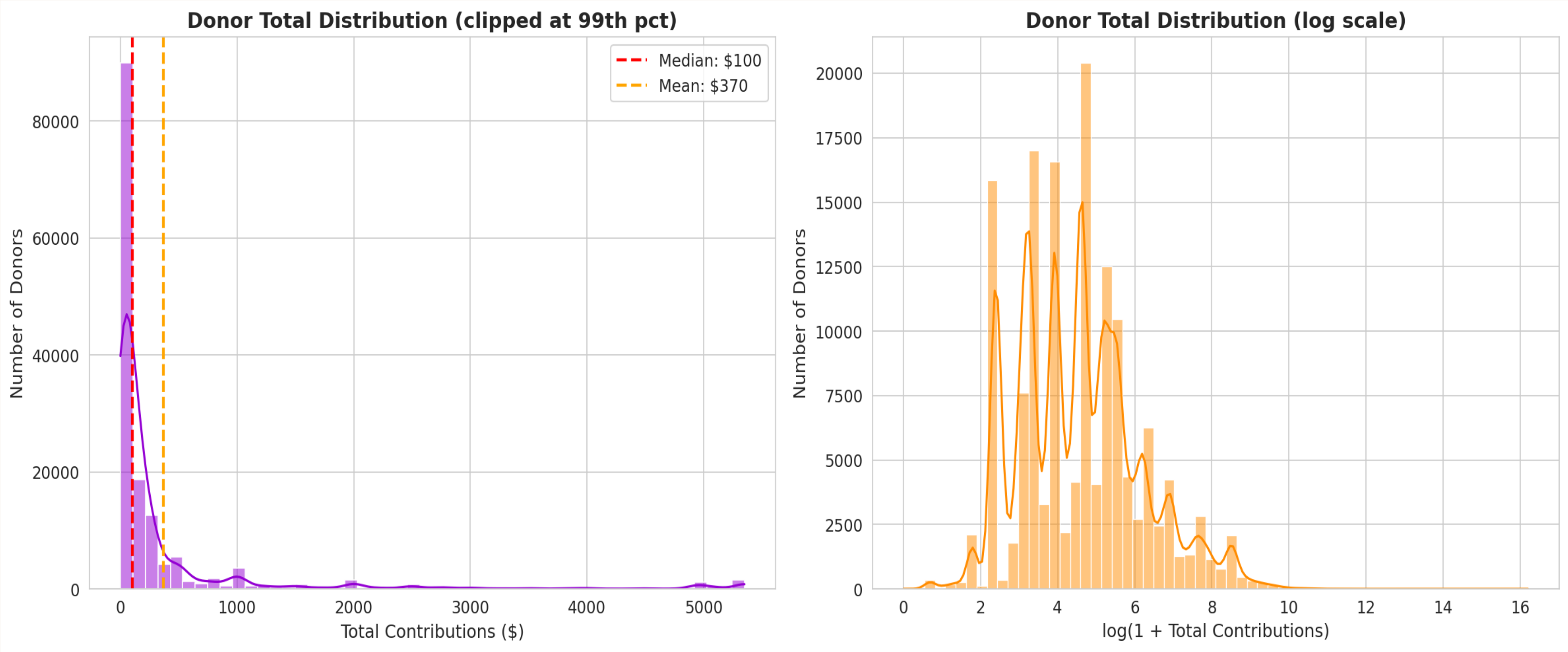
Seaborn histplot + KDE reveals a strongly right-skewed transaction distribution. The log-transformed view normalizes it — justifying the log(1+x) target transformation for ML modeling.



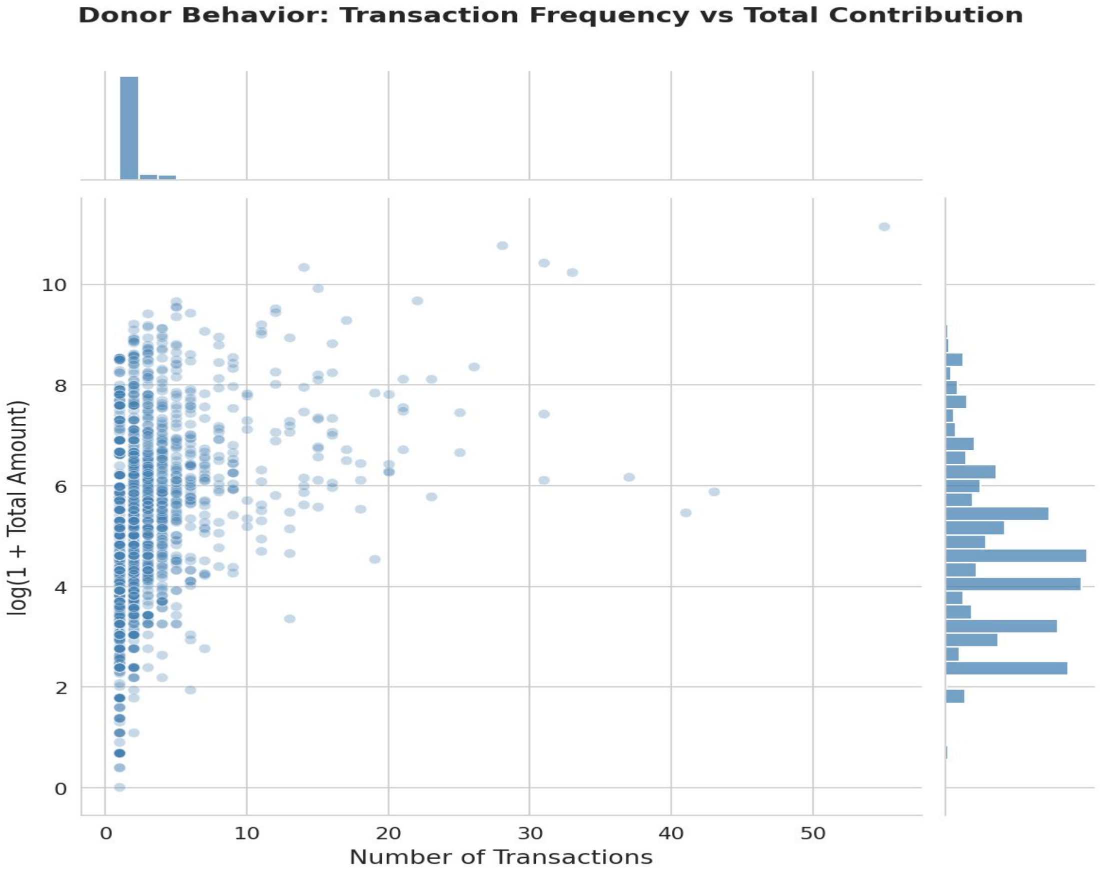


DONOR TOTAL DISTRIBUTION

Median donor total is 00 while the mean is 99 — a 5× gap driven by top donors. The dark-violet histogram highlights extreme concentration at the low end; the log-scale view reveals the full spread.



DONOR BEHAVIOR: TRANSACTION FREQUENCY VS TOTAL VALUE



Frequency ≠ Value

Many donors make 3–5 transactions yet have low totals. Repeat count alone is a poor predictor — amounts matter far more.

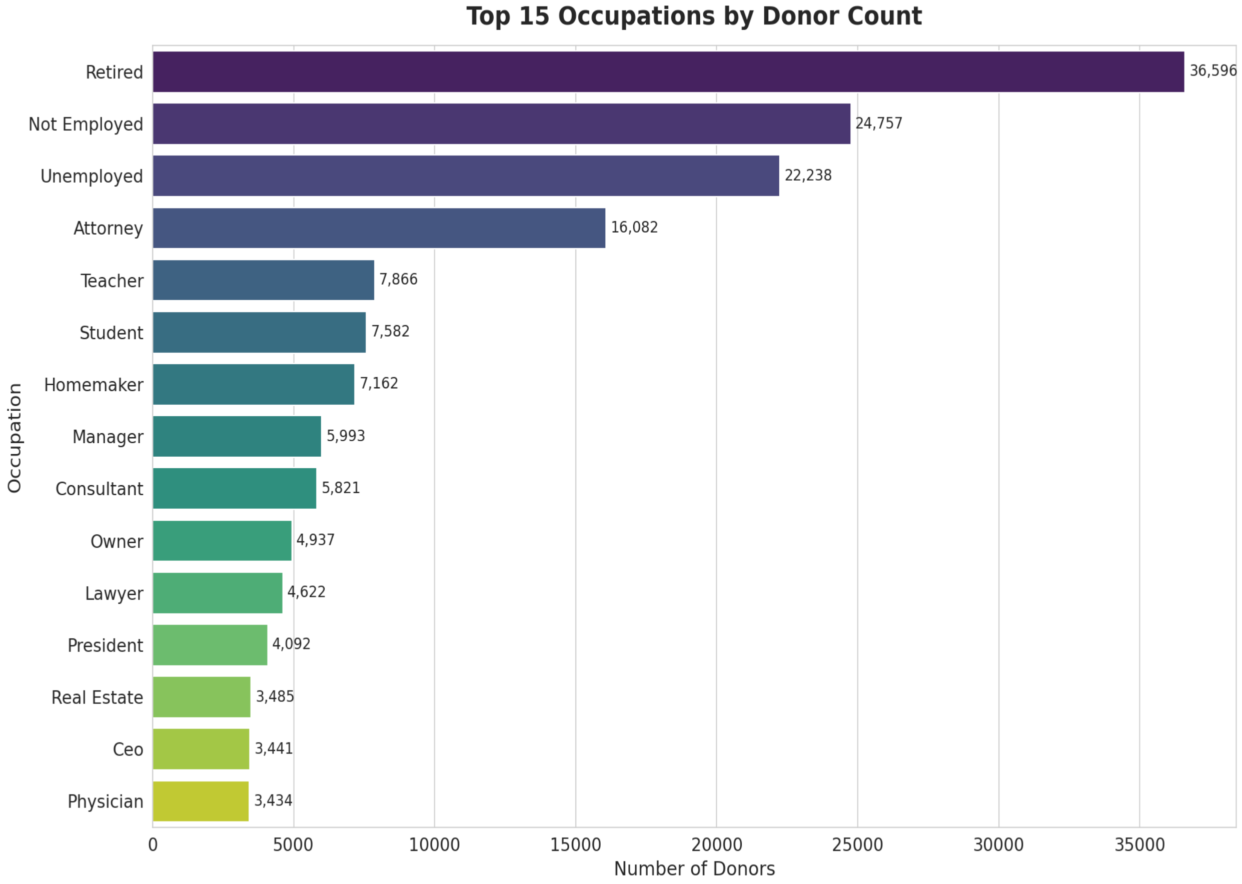
Power-Law Pattern

A small cluster of high-frequency donors (50+ transactions) concentrates upper-right: both frequent AND high-value.

Single-Gift Dominance

The dense column at x=1 confirms 72.6% of donors give once. The marginal histograms make this immediately visible.

WHO DONATES MOST — TOP 15 OCCUPATIONS BY DONOR COUNT



Key Takeaways

36,596

Retired donors — #1 by volume

16,082

Attorney donors — #4 but high value

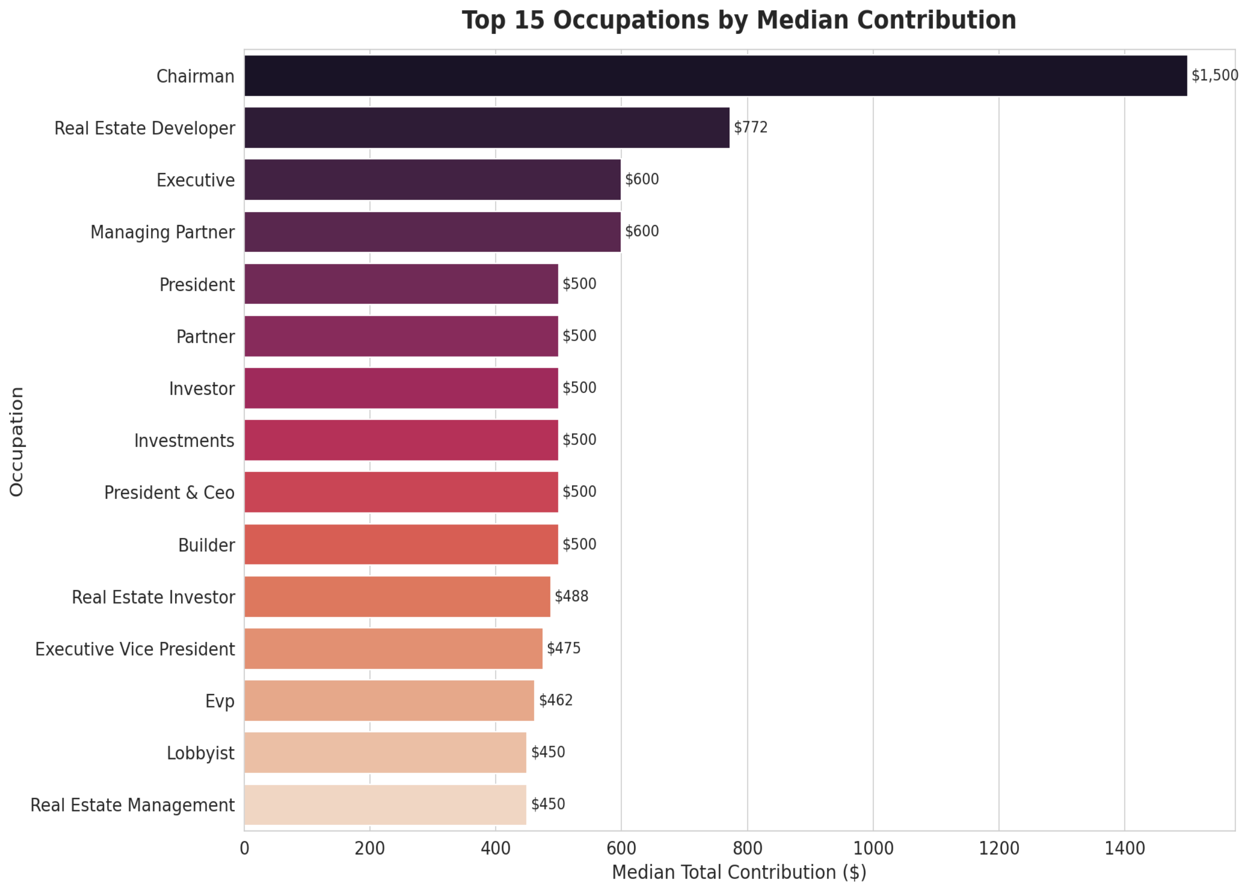
22,238

Unemployed donors — 3rd largest group

Insight

Volume ≠ Value. Retired & unemployed donors dominate count but give far smaller median amounts. Attorneys are 4th by count yet median giving is 6× higher.

HIGH-VALUE DONORS — TOP 15 OCCUPATIONS BY MEDIAN CONTRIBUTION



Key Takeaways

$1,500

Chairman median — highest of any occupation

$773

Real Estate Developer — 2nd highest median

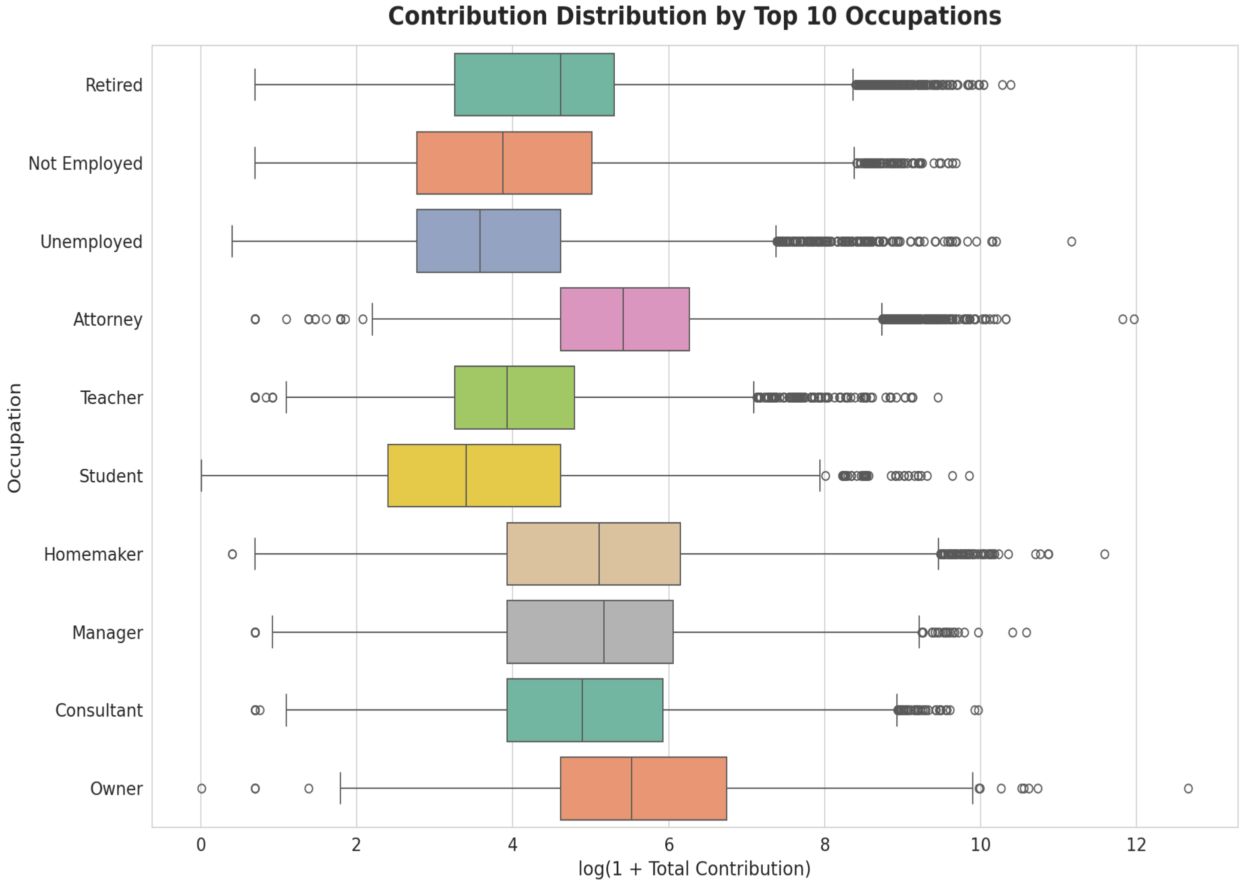
6×

Chairman gives vs. an average Individual

Insight

Finance, real estate, and executive titles cluster at the top. Priority segments for high-touch, high-dollar outreach campaigns.

GIVING SPREAD — CONTRIBUTION DISTRIBUTION BY OCCUPATION (BOXPLOT)



How to Read This

Box center

Median total contribution per donor in that occupation

Box width (IQR)

Middle 50% of donors — tighter = more consistent givers

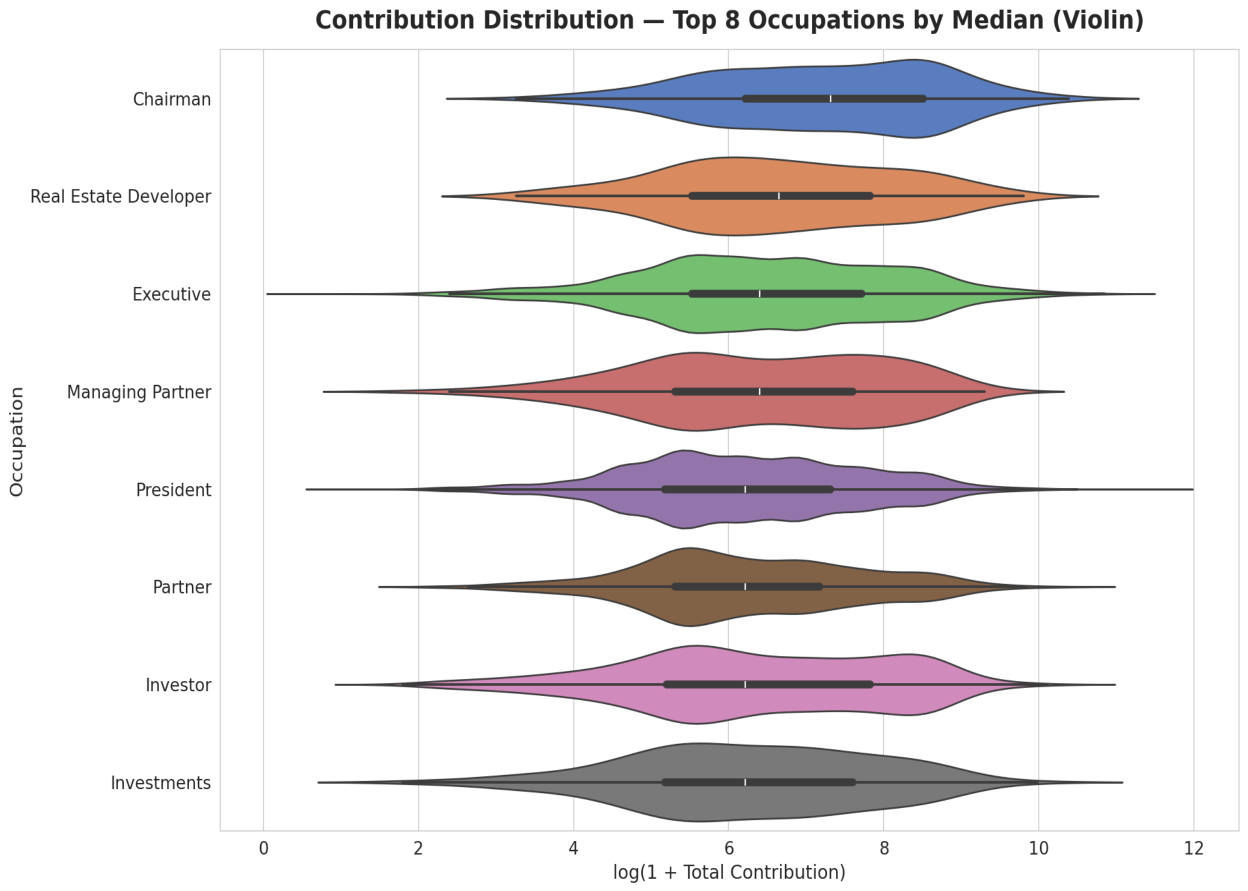
Whiskers & dots

Outlier donors — the extreme high-value individuals in each group

Insight

Attorneys have the widest IQR — highly variable, from small givers to major donors. Owners and Consultants show tighter, more predictable distributions.

GIVING SHAPE — FULL DISTRIBUTION BY OCCUPATION (VIOLIN PLOT)



How to Read This

Width of violin

Density of donors at that level — wider = more donors giving that amount

Bimodal shapes

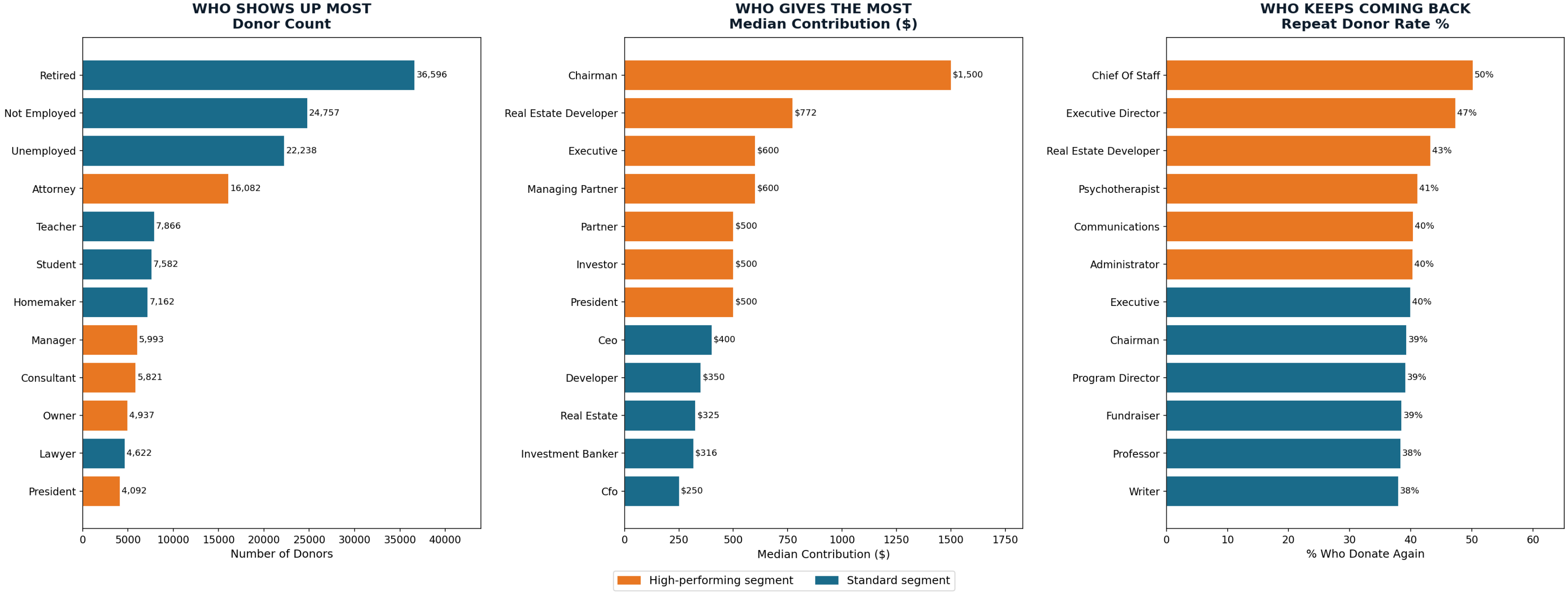
Two humps = two distinct donor clusters within the same occupation

Long right tails

Thin extension to the right signals outlier whale donors in that group

Insight
Chairman and Real Estate Developer show the most pronounced right skew. Executives show a notably uniform spread — predictable high-value donors.

WHAT OCCUPATIONS DONATE — THREE WAYS TO LOOK AT IT

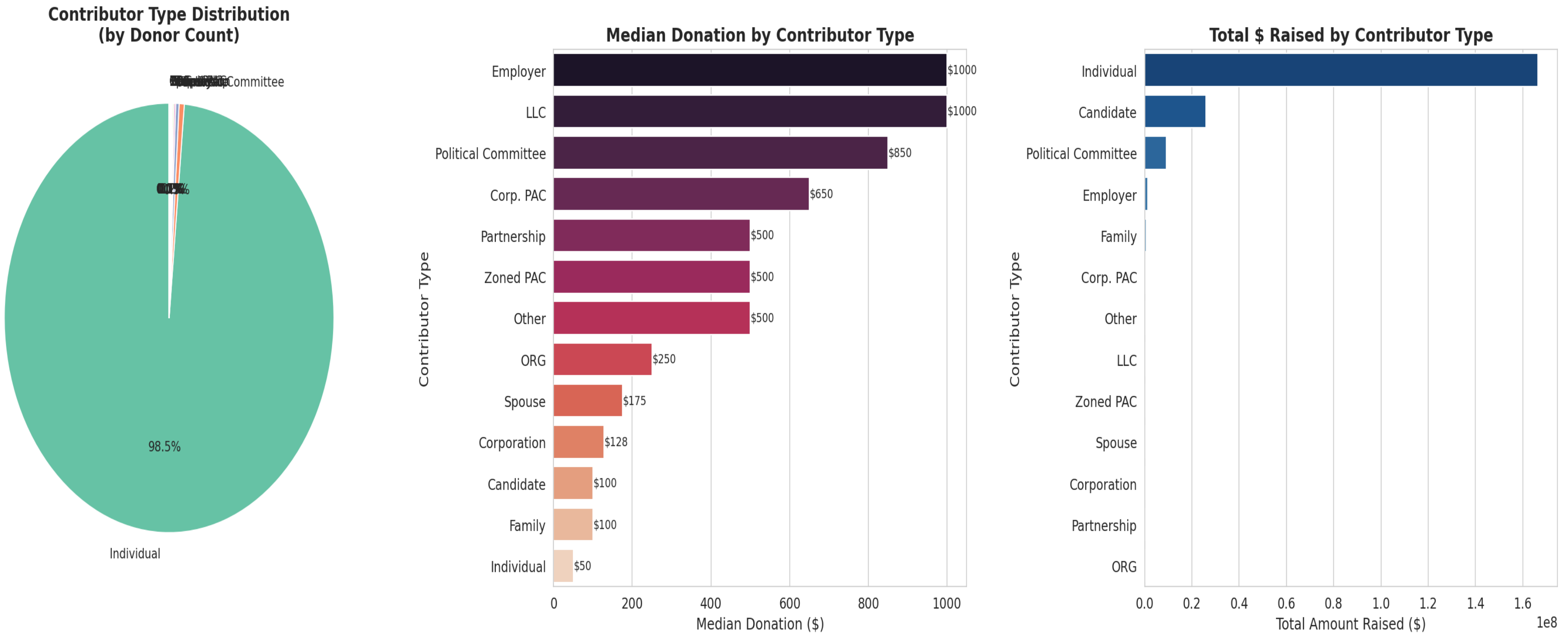


The Core Tension:

The occupations that show up most (Retired, Unemployed) give the least and rarely return. The occupations that give the most and keep coming back (Chairmen, Real Estate Developers, Executive Directors) are a small but enormously valuable minority.

WHO CONTRIBUTES — CONTRIBUTOR TYPE BREAKDOWN

98.5% of contributions come from individuals, but political committees, LLCs, and employer-type donors give significantly higher median amounts per transaction — critical for capacity modeling.



GEOGRAPHIC BREAKDOWN — CONTRIBUTIONS BY NYC BOROUGH

Borough Insights

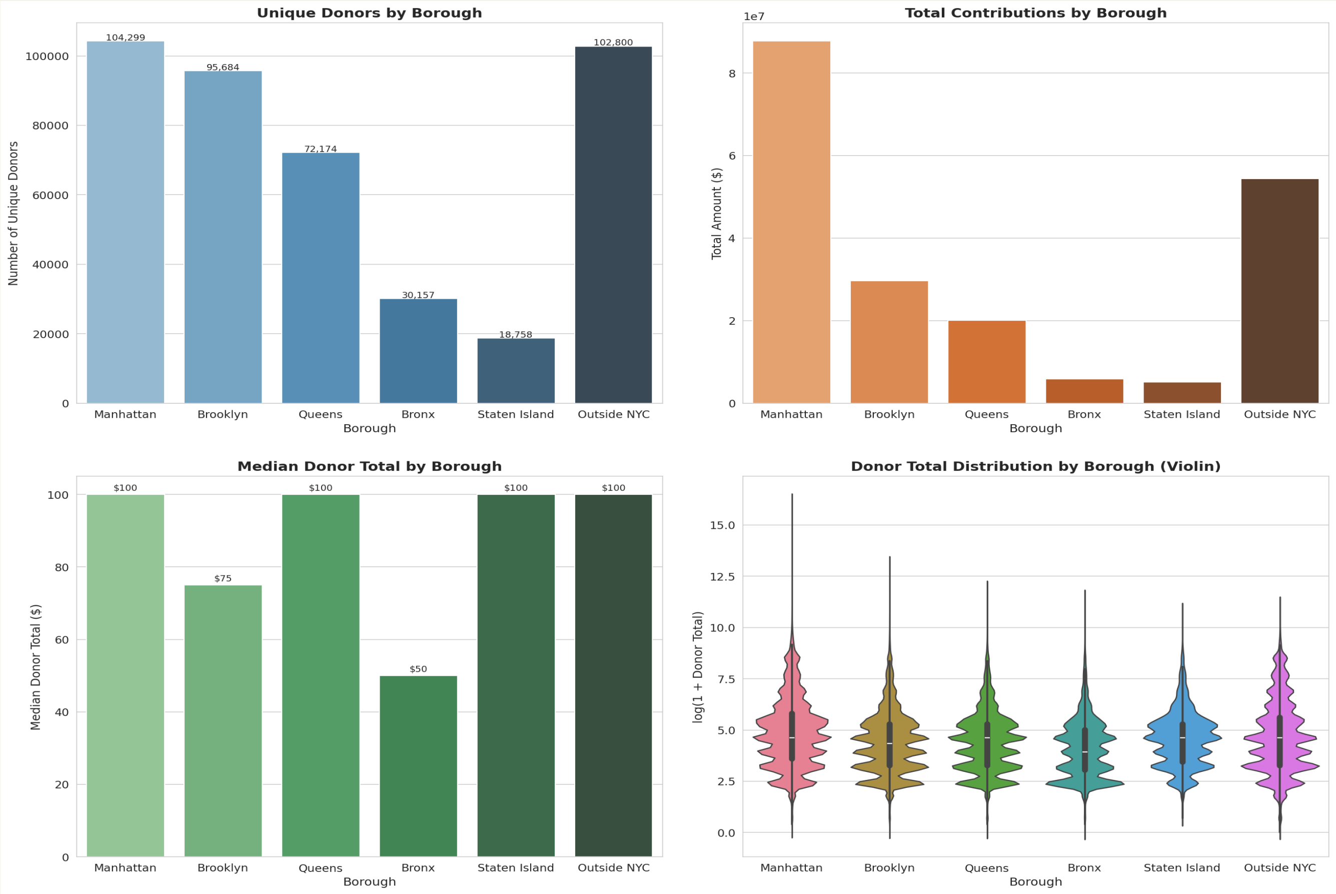
Manhattan:
Highest median giving — home to finance & law professionals

Brooklyn:
Largest donor count after Manhattan — diverse grassroots base

Outside NYC:
Significant out-of-state bundled donations via intermediaries

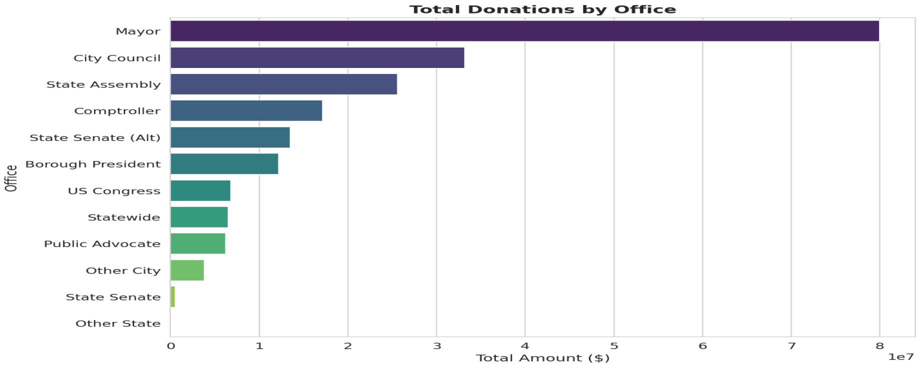
Bronx:
Lowest per-donor totals — community-level grassroots giving

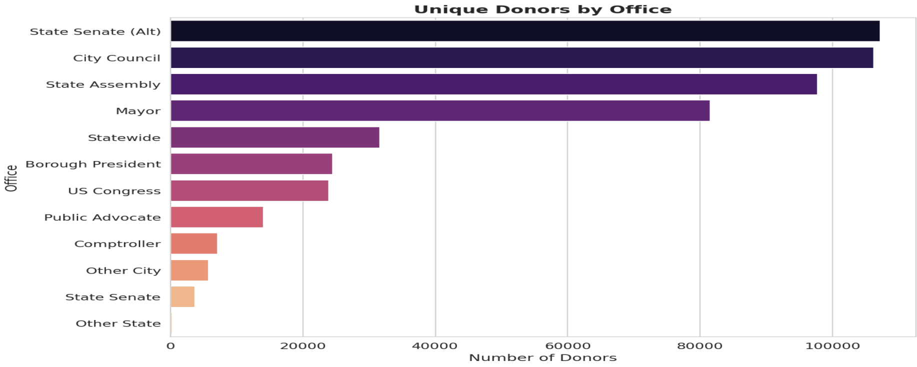
Staten Island:
Smallest base but above-average per-donor amounts

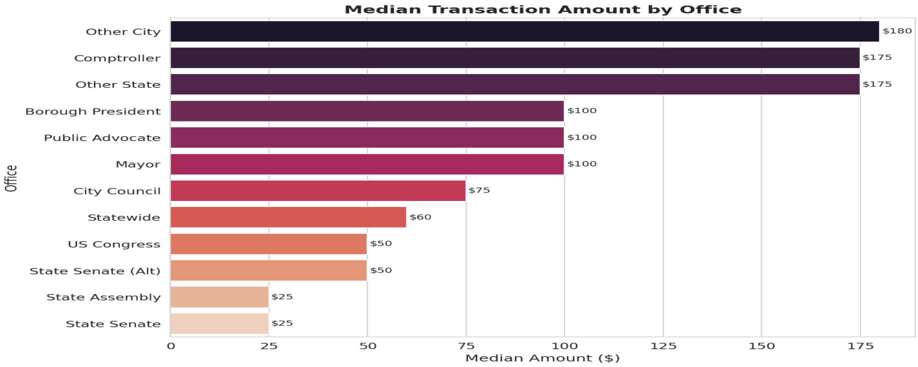


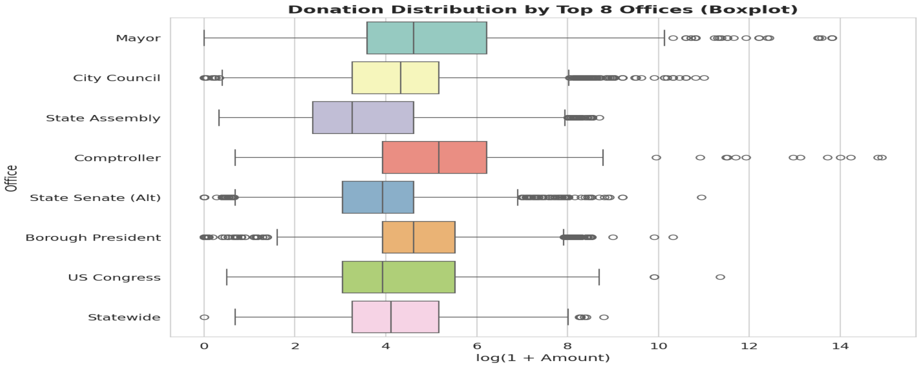
WHICH OFFICES ATTRACT WHICH DONORS — OFFICE CODE ANALYSIS

Mayoral and City Council races dominate by total volume, but Statewide and State Senate races command higher median transaction amounts — suggesting more organized, high-dollar fundraising at higher ballot positions.



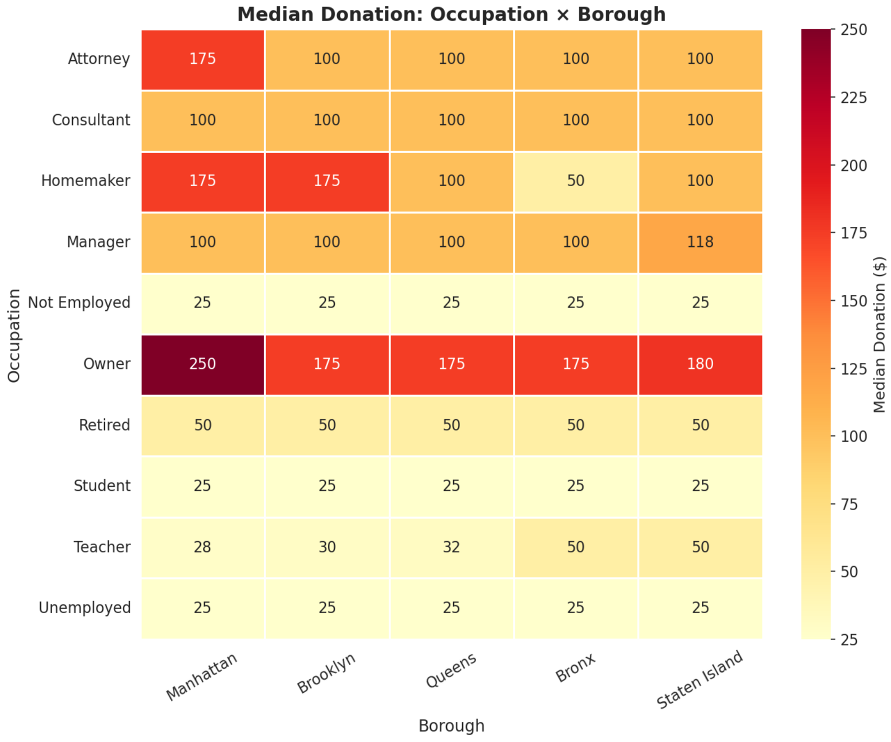


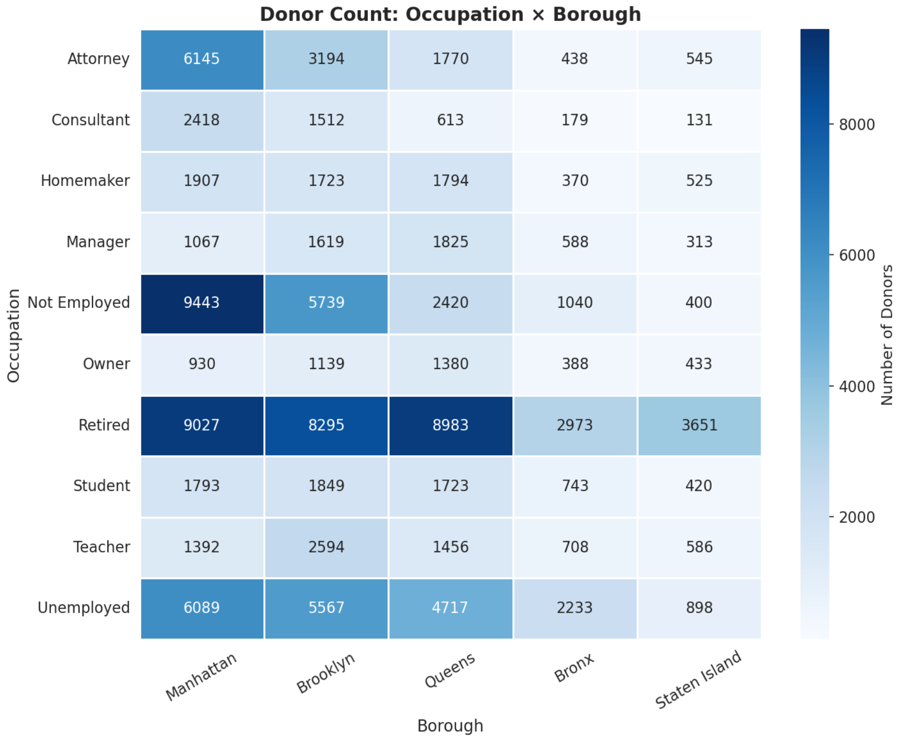




OCCUPATION × BOROUGH HEATMAP — MEDIAN DONATION & DONOR COUNT

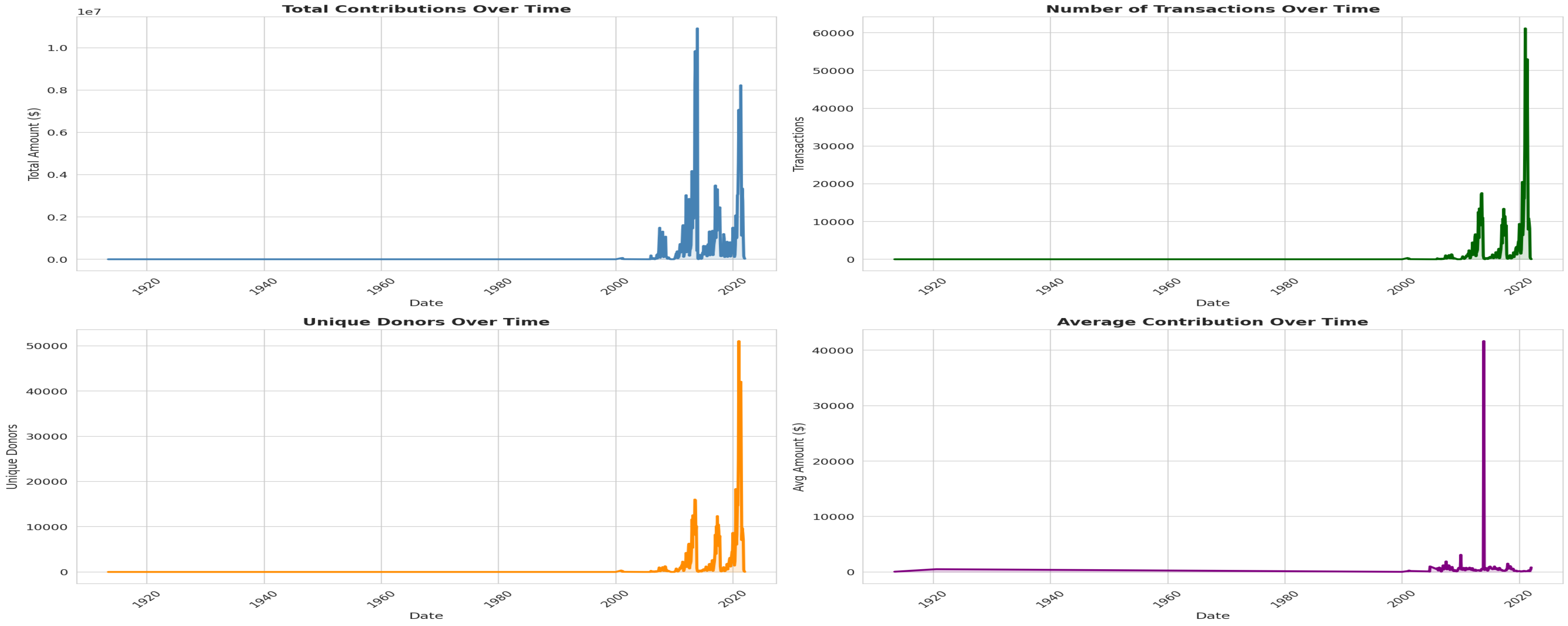
Cross-tabulating occupation with borough reveals where high-value donor segments cluster. Manhattan attorneys and owners give 2–4× more than their outer-borough counterparts — a powerful targeting signal for campaign outreach.



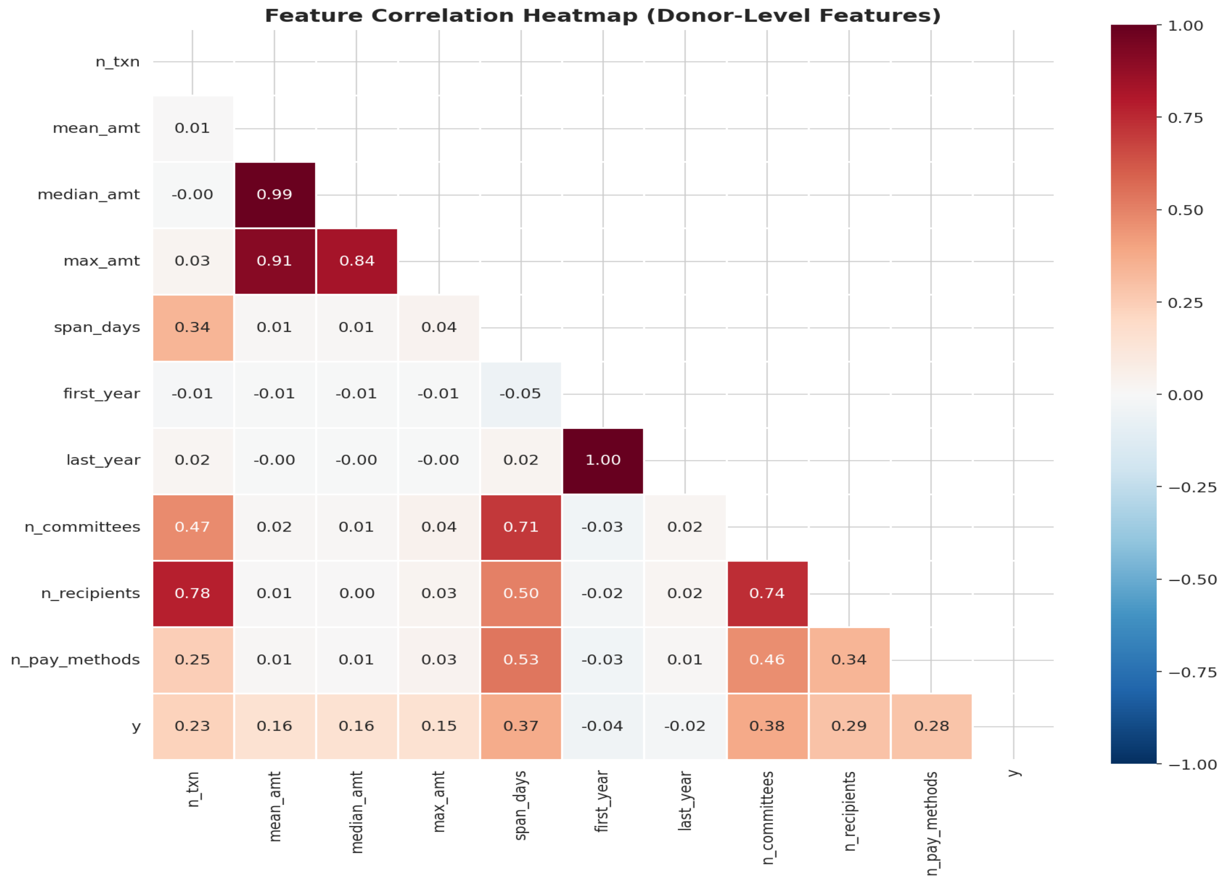


CAMPAIGN FINANCE TIME SERIES — MONTHLY TRENDS (2013–2021)

Seaborn lineplots with fill reveal seasonal election-cycle surges. Donations spike in the months before each election year then drop sharply post-election day. 2021 shows the largest surge on record.



FEATURE CORRELATION HEATMAP



Key Correlations

median_amt → y ~0.93

Strongest signal by far

mean_amt → y ~0.88

Confirms amount > frequency

max_amt → y ~0.70

Max gift cap matters

n_txn → y ~0.30

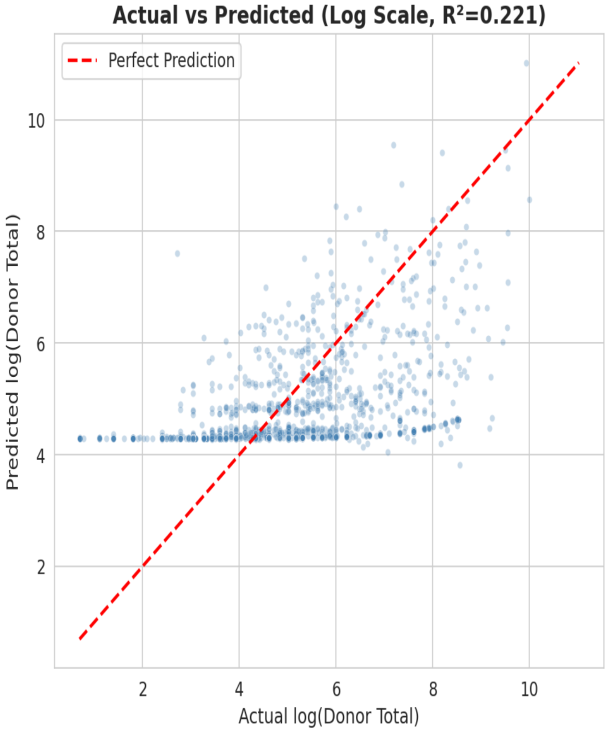
Frequency is weak alone

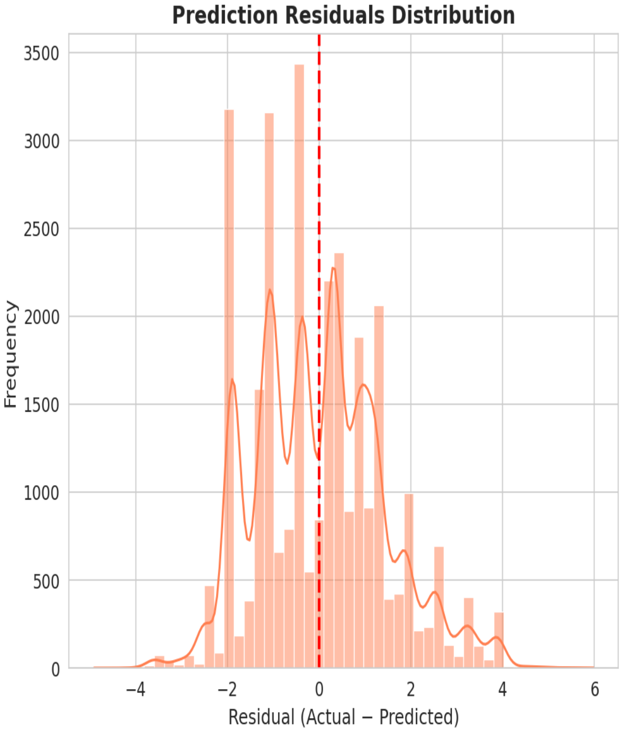
span_days → y ~0.25

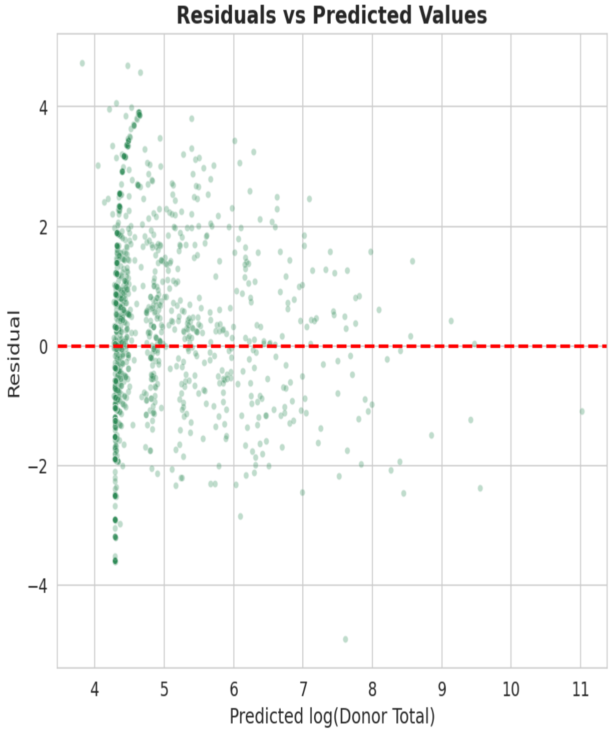
Tenure adds small signal

MODEL DIAGNOSTICS — PREDICTED vs ACTUAL · RESIDUALS

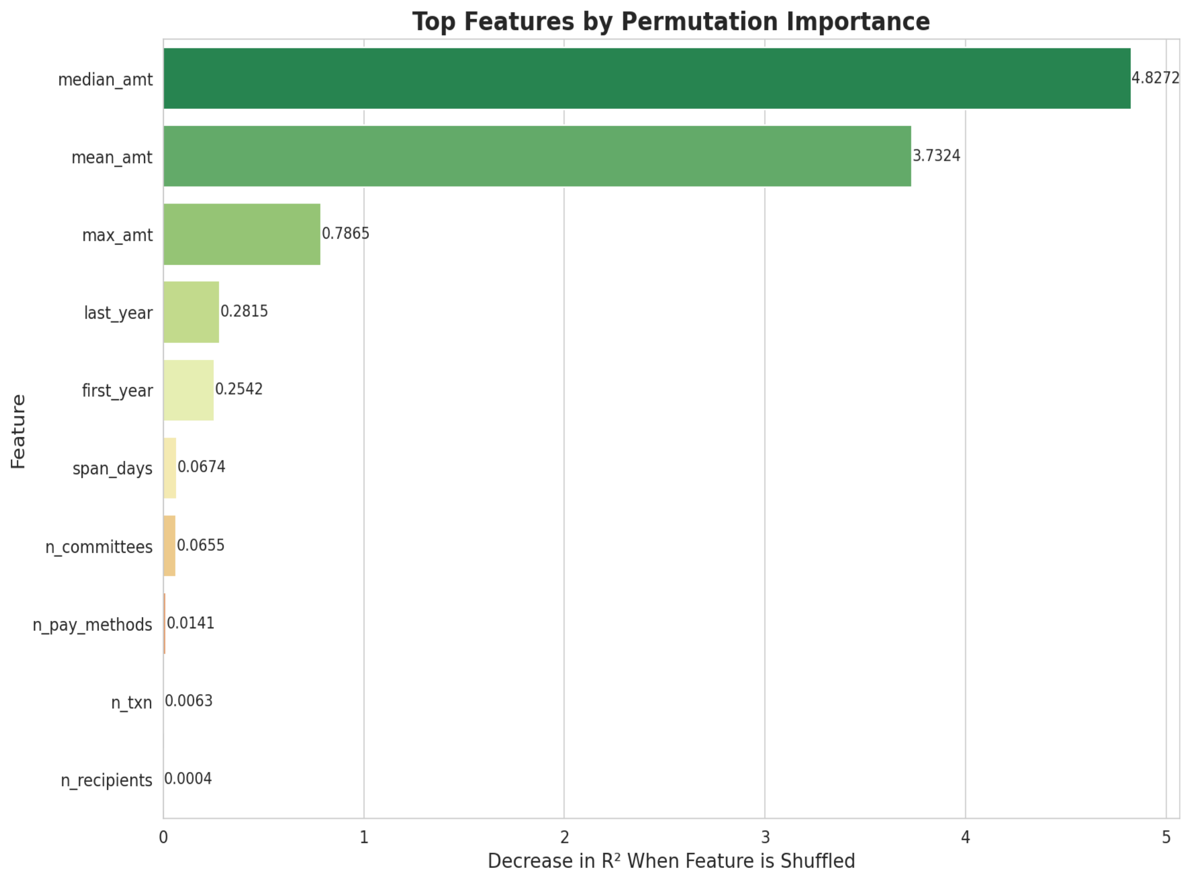
Three diagnostic panels: the scatter closely tracks the perfect-prediction line for mid-range donors; residuals are near-normally distributed; and the residual-vs-predicted plot shows no major heteroskedasticity.







PERMUTATION FEATURE IMPORTANCE (RdYlGn Palette)

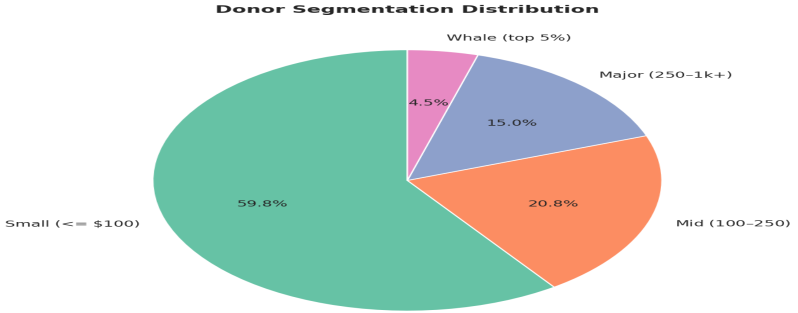


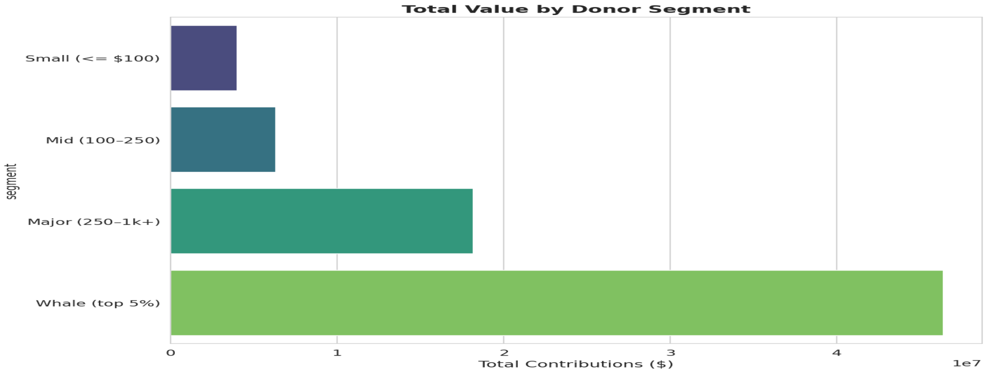
Interpretation

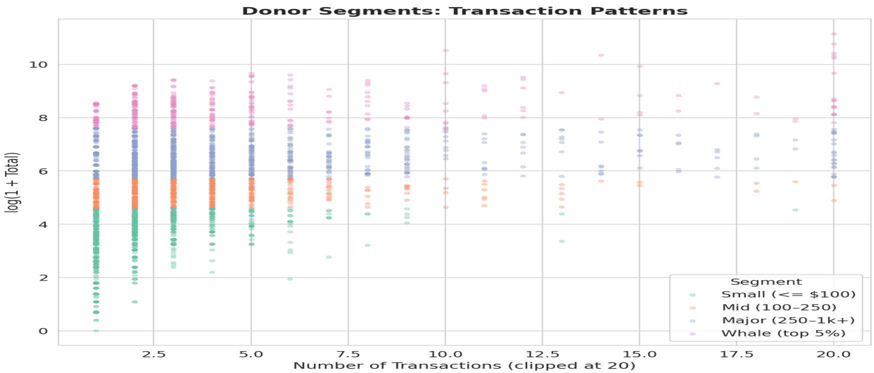
- median_amt (4.83) & mean_amt (3.73) dominate — 10–60× more predictive than any other feature
- max_amt (0.79) adds value — a single large gift signals capacity
- last_year / first_year provide modest signal — recency matters some
- span_days, n_committees add minimal value after amounts are accounted for
- n_txn is near zero — frequency is a lagging, not leading, indicator

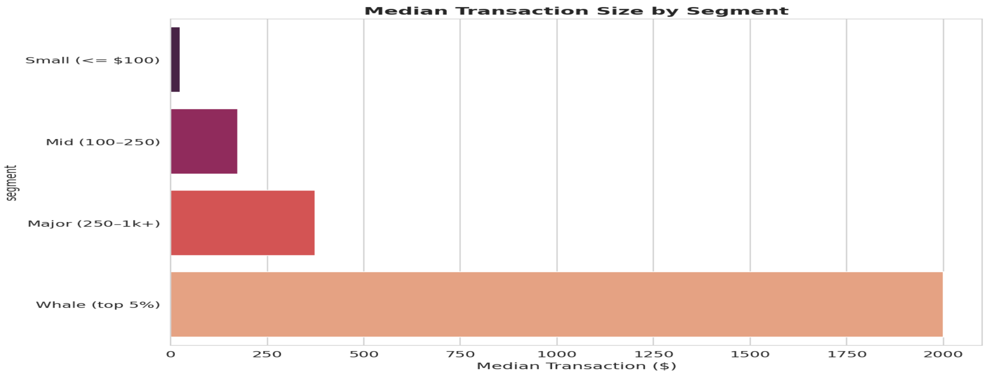
DONOR SEGMENTATION — PIE · VALUE · SCATTER · MEDIAN TXN

Donors segmented into four tiers by total giving. Whales (top 5%) represent a tiny fraction of donors but an outsized share of total contributions. The scatter confirms whales are also frequent transactors.









STRATEGIC RECOMMENDATIONS

Turning ML insights into actionable campaign strategy

01

Prioritize Upgrade Campaigns for Single-Gift Donors

72.6% give only once. A targeted second-gift ask within 90 days could significantly boost lifetime value across the base.

02

Score Donors by Median Gift Size, Not Frequency

The model and correlation heatmap confirm median & mean gift amount are the strongest predictors. Past giving level >> frequency.

03

Target Manhattan Finance & Real Estate Professionals

The occupation × borough heatmap shows Owners and Attorneys in Manhattan give 2–4× the borough average. Highest ROI outreach segment.

04

Build a Whale Classifier Before Full Regression

A top-5% binary classifier followed by regression would dramatically improve accuracy for the highest-impact donor segment.

NEXT STEPS & MODEL ROADMAP

SHAP Explainability:	Add SHAP values for local, per-donor prediction explanations to build campaign team trust in the model.
First-Touch Scoring:	Predict future value using only the first donation — actionable within 24 hours of donor acquisition.
Whale Classifier:	Train a binary classifier to identify top 5% donors early, enabling premium outreach before competitors.
Time-Horizon Prediction:	Predict '90-day value' or 'this-cycle total' rather than all-time total for operationally useful scoring.
Real-time API:	Wrap the model in a REST endpoint to score new donors in real-time as they enter the CRM.